

SIMATS ENGINEERING

ASSESSMENT TOOL III – ERROR ANALYSIS TASK (LOW)

Course Name: Ethical Hacking	Code: ITA1416	CO: CO2
Duration: 90 min Date:	Total Marks:	Reg. No / Name : 192325090 / K Nitheesh Kumar

CO2: Execute footprinting in social engineering such as DNS analysis, competitive intelligence gathering, and attack simulations to collect relevant information for identifying potential security risks. (BL3)

Assessment Tool Weightage: 35%

Instructions: Answer all questions. Each question carries 10 marks.

1. A student performs footprinting using Google dorks but constructs queries without operators like “site:”, “filetype:”, or quotation marks. As a result, the search results include unrelated websites and irrelevant documents. The student proceeds with analysis using this incorrect data. Identify the error in query formulation and explain why improper search syntax leads to inaccurate footprinting. Demonstrate how to correct the search query using proper operators and refine results for accurate intelligence gathering.

The student's mistake is the incorrect formulation of Google search queries by not using Google Dork operators such as site:, filetype:, intitle:, inurl:, or quotation marks (""). Without these operators, Google performs a broad search and returns many unrelated or irrelevant results, making the footprinting process inaccurate and unreliable.

Improper search syntax leads to inaccurate footprinting because:

- It retrieves unrelated websites and documents.
site:example.com filetype:pdf "annual report"

Error in Google Dork Query Formulation and Correct Footprinting Method

Footprinting is the first phase of ethical hacking and cybersecurity assessment, where publicly available information about a target organization is collected. One of the most effective

footprinting techniques is **Google Dorking (Google Hacking)**, which uses advanced Google search operators to locate specific information indexed by search engines. However, the accuracy of the collected information depends entirely on how the search query is constructed.

In the given scenario, the student performs footprinting using Google dorks but **fails to use essential search operators** such as site:, filetype:, intitle:, inurl:, and quotation marks ("").

Instead, the student enters general keywords into Google. As a result, Google searches the entire web and returns thousands of unrelated websites, blogs, advertisements, and irrelevant documents. The student then analyzes this incorrect information, leading to inaccurate conclusions and ineffective intelligence gathering. This is a major mistake because footprinting relies on collecting **precise and relevant information**, not random search results.

Improper search syntax reduces the effectiveness of footprinting because search engines interpret broad keywords in many different ways. Without search operators, results may include content from unrelated domains, duplicate information, outdated documents, or websites that have no connection to the target organization. Consequently, the collected intelligence becomes unreliable, wastes investigation time, and may cause security analysts to overlook valuable information such as exposed documents, login pages, configuration files, or publicly accessible reports.

The correct approach is to refine the search using **Google Dork operators**, which limit the search to specific websites, file types, or exact phrases. Some examples are:

- site:example.com – Searches only within the target domain.
- filetype:pdf site:example.com – Finds PDF documents published by the organization.
- site:example.com "confidential" – Searches for pages containing the exact word "**confidential**" within the target website.
- intitle:"index of" site:example.com – Searches for directory listings.
- inurl:login site:example.com – Finds login pages within the target domain.

For example:

Incorrect Query:

company annual report confidential

This query searches the entire internet and returns many unrelated results.

Correct Query:

site:company.com filetype:pdf "annual report"

This query searches **only the company's website**, limits the results to **PDF files**, and finds documents containing the **exact phrase "annual report."** The search results become highly relevant, making the footprinting process more accurate and efficient.

In conclusion, using proper Google Dork syntax is essential for successful footprinting. Search operators filter unnecessary information, improve search accuracy, reduce false positives, and

help security professionals gather reliable intelligence from publicly available sources. Ethical hackers should always use precise queries and analyze only relevant data while ensuring that all information gathering is conducted legally and ethically.

2. While conducting competitive intelligence, a student gathers company data from blogs and unofficial forums without verifying authenticity. The collected information includes outdated employee details and incorrect organizational structure. Identify the mistake in source validation and explain why unreliable data affects decision-making. Recommend methods to verify information using official sources such as company websites, LinkedIn, and government databases.

Mistake in Source Validation and the Importance of Using Reliable Sources in Competitive Intelligence

Competitive intelligence is the process of collecting and analyzing publicly available information about an organization, its competitors, products, employees, and business strategies. The information gathered should be accurate, current, and obtained from legal and ethical sources. In the given scenario, the student collects company information from **blogs and unofficial online forums** without verifying whether the information is authentic. As a result, the collected data contains outdated employee details and an incorrect organizational structure. This is a major mistake because the student has failed to **validate the credibility and reliability of the information sources** before using them for analysis.

The primary error is relying on **unofficial and unverified sources**. Blogs, discussion forums, and social media posts may contain personal opinions, rumors, outdated content, or incorrect information that has not been reviewed or confirmed by the company. Such sources may not reflect recent organizational changes, employee movements, or business updates. Using inaccurate information can lead to incorrect conclusions, poor business decisions, ineffective security assessments, and unreliable competitive analysis. In cybersecurity and business intelligence, decisions based on false information may cause organizations to misidentify competitors, overlook actual risks, or waste resources investigating incorrect data.

To ensure the accuracy of competitive intelligence, the collected information should always be verified using **official and trusted sources**. The following methods can be used:

1. Company Website

The official company website is the most reliable source for information about:

- Company profile and history
- Products and services
- Leadership team and management

- Press releases and announcements
- Organizational updates and contact details

2. LinkedIn

LinkedIn provides updated professional information such as:

- Current employees and job roles
- Organizational hierarchy
- Company size and locations
- Employee experience and skills
- Recent hiring and career updates

3. Government Databases

Government and regulatory databases provide legally verified business information, including:

- Company registration details
- Business licenses
- Annual financial reports
- Directors and legal representatives
- Regulatory compliance information

The student's mistake is using unverified and unofficial sources such as blogs and online forums without checking whether the information is accurate, current, or trustworthy. Competitive intelligence should always rely on credible and verified sources to ensure the collected information is reliable.

Using unreliable data negatively affects decision-making because:

- It may contain outdated or incorrect information.
- It can lead to inaccurate analysis of the organization.
- It may result in poor business or security decisions.
- It wastes time and resources by acting on false information.
- It reduces the credibility of the intelligence gathered.

To verify the authenticity of information, the following official sources should be used:

- **LinkedIn:** Confirm employee names, job titles, organizational structure, and company updates.
- **Government Databases:** Check company registration details, legal status, business licenses, and regulatory information.
- **Official Press Releases:** Verify recent company announcements and business developments.
- **Annual Reports and SEC/Corporate Filings (where applicable):** Confirm financial information, executive management, and organizational structure.

By collecting information only from **official and trusted sources** and cross-verifying it across multiple platforms, competitive intelligence becomes more accurate, reliable, and useful for informed decision-making.

3. A student directly starts active scanning on a target organization without performing passive footprinting such as WHOIS lookup and search engine analysis. This results in detection alerts being triggered. Identify the mistake in reconnaissance methodology and explain why passive reconnaissance is critical. Suggest the correct sequence of footprinting steps.

The student's mistake is starting active reconnaissance before completing passive footprinting. Active scanning directly interacts with the target's systems, making it more likely to be detected by firewalls, intrusion detection systems (IDS), or intrusion prevention systems (IPS). As a result, security alerts are triggered, revealing the presence of the reconnaissance activity.

Passive reconnaissance is critical because:

- It gathers information without directly interacting with the target's systems.
- It minimizes the risk of detection and alert generation.
- It provides valuable information such as domain names, IP addresses, DNS records, employee details, and organizational information.
- It helps plan targeted and efficient active scanning.
- It reduces unnecessary network traffic and avoids early exposure.

Correct Sequence of Footprinting Steps:

- 1. Identify the Target Organization** – Define the scope of the reconnaissance.

2. **Perform Passive Footprinting** – Collect publicly available information using:

- WHOIS lookup

Analyze the Collected Information – Identify domains, IP addresses, subdomains, employees, and network details.

3. **Perform Active Scanning** – Conduct ping sweeps, port scanning, DNS enumeration, and service detection only after passive reconnaissance is complete.
4. **Document and Verify Findings** – Validate the collected information and prepare the final intelligence report.

Following this sequence ensures that reconnaissance is systematic, accurate, and less likely to be detected, improving the effectiveness of the footprinting process.

4. A student performs WHOIS lookup but only records domain name and registrar details, ignoring expiry date, administrative contacts, and name servers. Identify the missing components and explain how these fields provide useful security insights. Suggest how proper WHOIS analysis can reveal vulnerabilities.

The student's mistake is recording only the domain name and registrar information while ignoring other important WHOIS fields. A complete WHOIS analysis includes several additional components that provide valuable information for security assessment and footprinting.

Missing Components:

- Domain Expiry Date
- Administrative Contact Information
- Technical Contact Information
- Registrant Information (if publicly available)
- Name Servers (DNS Servers)
- Creation Date
- Last Updated Date
- Domain Status

Security Insights Provided by These Fields:

- **Expiry Date:** Indicates when the domain registration expires. An expired or soon-to-expire domain may be vulnerable to domain hijacking if not renewed.

- **Administrative Contact:** Provides details of the person or organization responsible for managing the domain, which can help identify key personnel during security assessments.
- **Technical Contact:** Identifies the technical administrator responsible for domain configuration and maintenance.
- **Name Servers:** Reveals the DNS servers managing the domain, helping analysts understand the DNS infrastructure and identify possible DNS misconfigurations.

- **Creation and Updated Dates:** Show the age of the domain and recent modifications, which can help detect newly registered or suspicious domains.
- **Domain Status:** Indicates whether the domain is active, locked, or under transfer, helping assess the domain's security configuration.

How Proper WHOIS Analysis Can Reveal Vulnerabilities:

A thorough WHOIS analysis can identify potential security weaknesses, such as:

- Domains that are close to expiration and susceptible to hijacking.
- Misconfigured or outdated DNS servers.
- Publicly exposed administrative or technical contact information that could be targeted in social engineering attacks.
- Recently registered or modified domains that may indicate suspicious activity.
- Weak domain management practices that could increase security risks.

By examining **all WHOIS fields**, security analysts gain a more comprehensive understanding of the target's domain infrastructure, enabling them to identify potential vulnerabilities and improve the accuracy of footprinting and reconnaissance.

5. A student downloads publicly available documents from an organization's website but focuses only on the visible content while ignoring embedded metadata. As a result, important details such as author names, software versions, and internal usernames are missed. Identify the mistake in the student's footprinting process and explain how metadata can provide valuable intelligence. Describe the correct method to extract metadata using appropriate tools. Suggest how this information could be used ethically to identify potential security risks.

The student's mistake is analyzing only the visible content of the downloaded documents while ignoring the embedded metadata. Metadata often contains hidden information that is not visible to users but can provide valuable intelligence during footprinting.

How Metadata Provides Valuable Intelligence:

Metadata can reveal important information such as:

- **Author names** of the document.
- **Software and application versions** used to create or edit the file.

- **Internal usernames** or computer names.
- **Document creation and modification dates.**
- **Company or organization details.**
- **File paths and document properties.**

This information helps security analysts understand the organization's environment, identify technologies in use, and detect potential security weaknesses.

Correct Method to Extract Metadata:

The metadata can be extracted using specialized metadata analysis tools such as:

- **ExifTool**
- **FOCA (Fingerprinting Organizations with Collected Archives)**
- **Metagoofil**
- **pdftinfo** (for PDF documents)
- **Microsoft Office Document Inspector** (for Office files)

Steps:

1. Download publicly available documents from the organization's website.
2. Open the document using a metadata extraction tool.
3. Extract and review metadata fields such as author, software version, creation date, modification date, and document properties.
4. Analyze the extracted information for security assessment and document the findings.

Ethical Use of Metadata Information:

The extracted metadata should only be used for **authorized security assessments** to identify and report potential security risks, such as:

- Outdated software versions that may contain known vulnerabilities.
- Exposure of internal usernames or computer names that could aid attackers.
- Disclosure of unnecessary document properties or file paths.

- Information leakage through hidden metadata.

The findings should be **reported responsibly to the organization** so that exposed metadata can be removed, sensitive information can be protected, and overall security can be improved.

Code of Conduct:

I 192325090 – Nitheesh Kumar K. certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Nitheesh Kumar K

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Error Identification	20%				
Root Cause Analysis	30%				
Correction & Justification	25%				
Application of Concepts	15%				
Clarity & Presentation	10%				

Criteria	Error Identification (2)	Root Cause Analysis (3)	Correction & Justification (2)	Application of Concepts (2)	Clarity & Presentation (1)	Total
Q1						
Q2						
Q3						
Q4						
Q5						

